Harikrishna Pillai

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Education

Technische Universität Dresden — Dresden, Germany
Master of Science in Computational Modelling and Simulation — *Oct 2025 – Present*

Amrita Vishwa Vidyapeetham — Kerala, India
B. Tech, Computer Science and Artificial Intelligence — *Oct 2021 – Aug 2025*

Experience

AI Research Intern — July 2025 – Present
Lossfunk — *Bangalore, India*

- Created a framework called MAYA that solves class-incremental learning without training and gradients. Achieved it by using geometrical manipulation of the trained models.
- MAYA uses brain like efficient episodic memory to perform very close to offline training.
- Building a Jax based deep learning framework that can efficiently use evolved dendritic neurons with various functions.

AI Research Intern (Homeostatic RL) — Nov. 2024 – Oct. 2025
CMC Lab (TU Dresden) — *Dresden, Germany*

- Worked on studying neural activity of homeostatic RL agents to find interesting patterns with its foraging behaviour.
- Developing a weight perturbation algorithm that achieves convergence comparable to Gradient Descent.
- The results are promising and comparable to Gradient Descent.

ML Engineer — Jan 2024 – June 2024
InSynk — *Remote*

- Worked on creating custom E-commerce agents like product background generation, apparel background change, AI Ad Generation, Product Description tool etc. Used by E-commerce companies like flipkart, myntra, big bazar, reliance and many more. **product link**
- Worked on creating python backend and REST APIs. Also creating scripts that could generate fastapi code dynamically. **product link**

ML Intern — Aug. 2023 – Dec. 2023
Aveta.ai — *Remote*

- Worked on making a special-purpose LLM for medical ICD coding.
- Used and explored techniques like RAG, Attention Manipulation and Prompt Engineering to optimize the LLM output.
- Created a tool that can analyze a PDF and intelligently separate all the patients data in different PDFs and save it.

Google Summer of Code 2022 @ INCF — June. 2022 – Sep. 2022
INCF — *Remote*

- Worked on mapping embryo microscopy images on a 3d ellipsoid generated using code. Created a Desktop application for its visualization.
- Used opencv library for image projection techniques and electron js for creating a desktop application to view and control the generated 3d object.

Research

MAYA: Manifold-Aligned Yielding Architecture for Continual Learning — Blog Link

- Designed and implemented MAYA, a gradient-free, analytic continual learning framework that overcomes the stability-plasticity dilemma by mitigating geometric bias in frozen neural networks.
- An ultra-efficient continual learning system for deploying large vision models (DINOv3, SigLIP2, ResNet50) on continuous data streams without catastrophic forgetting.

Homeostatic states shape neural and behavioral dynamics during foraging in embodied agents

- Worked to uncover the neural mechanisms underlying the observed behavior during foraging of Homeostatic RL agents.

CAN - Continuously Adapting Network Paper Link

- We introduced CAN, a continual learning approach that combines Hebbian learning, selective gradient freezing, and adaptive network expansion. CAN leverages Hebbian plasticity to identify task-relevant neurons and freeze them to prevent forgetting. For each new task, CAN selectively reuses or grows the network by duplicating highly active neurons, enabling scalability and adaptability without catastrophic forgetting.

Talks & Posters

Emergence of brain-like features in artificial neural networks trained based on homeostasis
(@ BrainBody 2025)

- Talk to uncover the neural mechanisms underlying the observed behavior during foraging of Homeostatic RL agents.

Brain-like features in homeostatic deep reinforcement learning agents
(@ Bernstein Conference 2025)

- Our model demonstrates the importance of homeostatic reward to study adaptive, survival-oriented behavior in ANNs. The observed coupling between nutrient levels, velocity, neural dynamics, and exploration patterns highlights how internal states modulate both cognition and neural dynamics.

Projects

insuranceLLM | *Python, ML, REST API* Aug 2023 – Dec 2023

- The goal of this project is to make an NLP model that can understand a doctor's clinical report and output the relevant ICD (International Disease Classification) codes.
- These codes are then used by the insurance company to give the insurance money.
- Challenging because we only have the ICD guidelines as the tabular data PDF.
- Worked on both vector similarity and Attention manipulation as the potential solutions.

Doodle Diffusion | *Python, Diffusion models*

- This project aims to build a diffusion based model that generates Google doodles from given prompts.
- The Reverse diffusion process is done using a UNet model which takes a noisy image, timestamp and text as inputs. The timestamp is embedded into the model using Sine Cosine Embedding and the text is encoded using a CLIP encoder and then added to the noisy image.

Professional References

Dr. Sushrut Thorat
Cognitive Computational Neuroscientist
https://www.sushrutthorat.com
sthorat@uos.de

Dr. Shervin Safavi Dresden, Germany
Principal Investigator @ CMC Lab
shervin.safavi@cmclab.org

Technical Skills

Languages: Python, C/C++, I just learn whatever is required
ML Frameworks & Libraries: Rest API, FastAPI, PyTorch, Jax, Django
Developer Tools: Git, Docker, Linux, VS Code, Blender, Godot
Libraries: MuJoCo, Isaaclab, Gym, OpenCV, numpy, pandas, Matplotlib